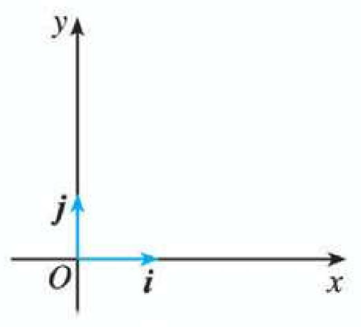
第十一章空间直角坐标系

(Space Rectangular Coordinate System)

一、知识引入(Knowledge Introduction)



平面直角坐标系

在平面几何中，我们通过平面直角坐标系将平面内的点和向量与有序实数对建立起一一对应关系，从而实现了几何问题的代数化求解。而在空间几何中，为了同样能够借助代数方法研究空间中的点、线、面关系，我们需要拓展平面直角坐标系的概念，建立空间直角坐标系，这是连接空间几何与代数运算的重要桥梁。(In plane geometry, we establish a one-to-one correspondence between points, vectors in the plane and ordered real number pairs through the plane rectangular coordinate system, thereby realizing the algebraic solution of geometric problems. In spatial geometry, to study the relationships(www.sicyes.com) between points, lines and planes in space using algebraic methods similarly, we need to extend the concept of the plane rectangular coordinate system and establish a space rectangular coordinate system, which is an important bridge connecting spatial geometry and algebraic operations.)

二、空间直角坐标系的建立(Establishment of Space Rectangular Coordinate System)

（一）核心定义(Core Definition)

在空间中选定一点 O 作为原点，再选定一个单位正交基底 $\{i,j,k\}$ （其中 i,j,k

是三个两两垂直且长度都为 1 的向量，称为坐标向量，满足$i \cdot j = j \cdot k = k \cdot i = 0$，$|i| = |j| = |k| = 1$）。以原点 O 为起点，分别以 i,j,k 的方向为正方向、以它们的长度为单位长度，建立三条互相垂直的数轴，依次称为 x 轴（横轴）、y 轴（纵轴）、z 轴（竖轴）。这三条坐标轴就构成了一个空间直角坐标系，记作 $Oxyz$ 。(In space, select a point O as the origin, and then choose an orthonormal basis $\mathbf{i},\mathbf{j},\mathbf{k}$ (where $\mathbf{i},\mathbf{j},\mathbf{k}$ are three vectors that are perpendicular to each other pairwise and have a length of 1. They are called coordinate vectors, satisfying $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$ and $|\mathbf{i}| = |\mathbf{j}| = |\mathbf{k}| = 1$).With the origin O as the starting point, take the directions of $\mathbf{i},\mathbf{j},\mathbf{k}$ as the positive directions and their lengths as the unit length to establish three mutually perpendicular number axes, which are called the x -axis (horizontal axis), y -axis (vertical axis) and z -axis (vertical axis) in sequence. These three coordinate axes form a space rectangular coordinate system, denoted as $Oxyz$.)

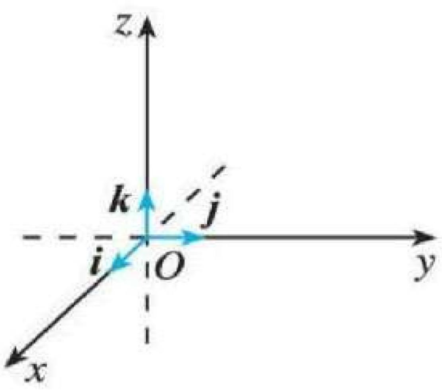
（二）相关概念(Related Concepts)

1. 坐标平面：通过每两条坐标轴的平面叫做坐标平面(www.sicyes.com)，空间直角坐标系中有三个坐标平面，分别是：(Coordinate Planes: The planes passing through each two coordinate axes are called coordinate planes. There are three coordinate planes in a space rectangular coordinate system, namely:)
2. Oxy 平面：由 x 轴和 y 轴确定的平面；(Oxy plane: the plane determined by the x -axis and y -axis;)
3. Oyz 平面：由 y 轴和 z 轴确定的平面；(Oyz plane: the plane

determined by the y -axis and z -axis;)

4. Ozx 平面：由 z 轴和 x 轴确定的平面。(Ozx plane: the plane determined by the z -axis and x -axis.)
5. 空间分区：三个坐标平面将整个空间分成八个部分，每个部分称为一个卦限。(Spatial Division: The three coordinate planes divide the entire space into eight parts, each called an octant.)

（三）坐标系的画法与类型(Drawing Methods and Types of Coordinate Systems)

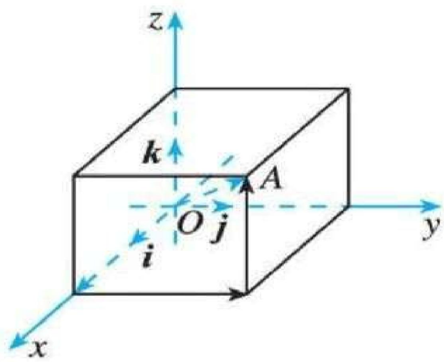


1. 绘制规则：画空间直角坐标系 $Oxyz$ 时，为了保证直观性，一般使 $\angle xOy = 135°$ （或 $45°$ ），$\angle yOz = 90°$ 。其中 x 轴和 y 轴通常画成水平方向， z 轴画成竖直方向。(Drawing Rules: When drawing the space rectangular coordinate system $Oxyz$, to ensure intuitiveness, generally make $\angle xOy = 135°$ (or $45°$) and $\angle yOz = 90°$. Among them, the x -axis and y -axis are usually drawn in the horizontal direction, and the z -axis is drawn in the vertical direction.)
2. 坐标系类型：本文中(www.sicyes.com)建立的坐标系均为右手直角坐标系。判断方法为：让右手拇指指向 x 轴的正方向，食指指向 y 轴的正方向，若中指能指向 z 轴的正方向，则该坐标系为右手直角坐标系。(Types of Coordinate Systems: All coordinate systems established in this book are right-handed rectangular coordinate systems. The judgment method is: let the thumb of the right hand point to the positive direction of the x -axis,

the index finger point to the positive direction of the y -axis, and if the middle finger can point to the positive direction of the z -axis, then the coordinate system is a right-handed rectangular coordinate system.)

三、空间中点与向量的坐标表示(Coordinate Representation of Points and Vectors in Space)

(一) 点的坐标表示(Coordinate Representation of Points)

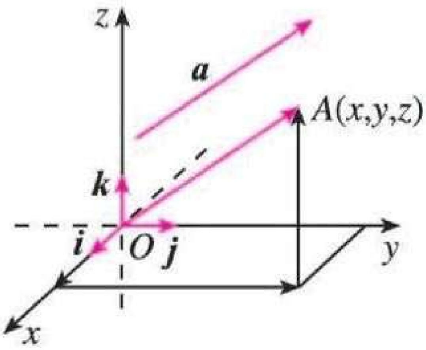


在空间直角坐标系 $Oxyz$ 中, 对空间任意一点 A , 都对应一个唯一的向量 $\overrightarrow{OA}$ (从原点 O 指向点 A 的向量)。根据空间向量基本定理, 存在唯一的有序实数组 (x,y,z) , 使得 $\overrightarrow{OA}=xi+yj+zk$ 。(In the space rectangular coordinate system $Oxyz$, for any point A in space, there corresponds a unique vector $\overrightarrow{OA}$ (the vector from the origin O to point A). According to the Fundamental Theorem of Spatial Vectors, there exists a unique ordered real number triple (x,y,z) such that $\overrightarrow{OA}=xi+yj+zk$.)

这个有序实数组 (x,y,z)就叫做点 A 在空间直角坐标系中的坐标, 记作 $A(x,y,z)$ 。其中: (This ordered real number triple (x,y,z) is called the coordinate of point A in the space rectangular coordinate system, denoted as $A(x,y,z)$. Among them:)

- x 叫做点 A 的横坐标; (x is called the abscissa of point A ;)
- y 叫做点 A 的纵坐标; (y is called the ordinate of point A ;)
- z 叫做点 A 的竖坐标。(z is called the z-coordinate of point A .)

（二）向量的坐标表示(Coordinate Representation of Vectors)



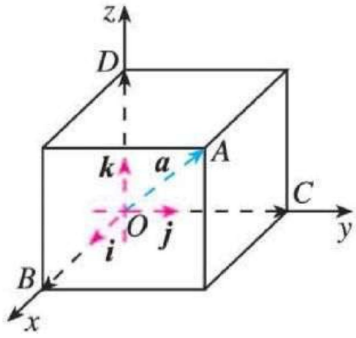
在空间直角坐标系 $Oxyz$ 中 (www.sicyes.com)，给定向量 $\mathbf{a}$ ，作 $\overrightarrow{OA}=\mathbf{a}$ （即把向量 $\mathbf{a}$ 的起点移至原点 O ），则向量 $\mathbf{a}$ 对应的终点 A 的坐标 (x,y,z) 就是向量 $\mathbf{a}$ 的坐标，记作 $\mathbf{a}=(x,y,z)$ ，且满足 $\mathbf{a}=xi+yj+zk$ 。(In the space rectangular coordinate system $Oxyz$, given a vector $\mathbf{a}$, construct $\overrightarrow{OA}=\mathbf{a}$ (that is, move the starting point of vector $\mathbf{a}$ to the origin O), then the coordinate (x,y,z) of the terminal point A corresponding to vector $\mathbf{a}$ is the coordinate of vector $\mathbf{a}$, denoted as $\mathbf{a}=(x,y,z)$, and it satisfies $\mathbf{a}=xi+yj+zk$.)

【注意】符号 (x,y,z) 具有双重意义：表示空间中的点A时，是点A的坐标；表示向量时，特指以原点O为起点、A为终点的位置向量$\overrightarrow{OA}$的坐标，自由向量的坐标需通过起点和终点坐标推导（如$\overrightarrow{AB}=(x_B-x_A, y_B-y_A, z_B-z_A)$）。

【Note】The symbol (x,y,z) has a dual meaning:When denoting a point A in space, it refers to the coordinates of point A ;When denoting a vector, it specifically refers to the coordinates of the position vector $\overrightarrow{OA}$ with the origin O as the starting point and A as the terminal point.The coordinates of a free vector need to be derived from the coordinates of its starting point and terminal point (e.g., $\overrightarrow{AB}=(x_B-x_A,y_B-y_A,z_B-z_A)$).

（三）坐标的几何意义与确定方法(Geometric Significance and Determination Method of Coordinates)

几何意义：过点 A 分别作垂直于 x 轴、y 轴和 z 轴的平面，依次交 x 轴、 y 轴和 z 轴于点 B 、 C 和 D 。则：(Geometric Significance: Pass through point A to make planes perpendicular to the x -axis, y -axis and z -axis respectively, intersecting the x -axis, y -axis and z -axis at points B , C and D in turn. Then:)



向量 $\overrightarrow{OA}$ 在 x 轴、y 轴、z 轴上的投影向量分别为 $\overrightarrow{OB}$ 、$\overrightarrow{OC}$ 、$\overrightarrow{OD}$ ；(The projection vectors of vector $\overrightarrow{OA}$ on the x -axis, y -axis and z -axis are $\overrightarrow{OB}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ respectively;)

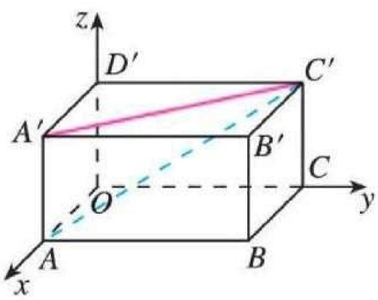
满足 $\overrightarrow{OA}=\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}$ ；(It satisfies $\overrightarrow{OA}=\overrightarrow{OB}+\overrightarrow{OC}+\overrightarrow{OD}$;)

点 B 、 C 、 D 在坐标轴上的坐标分别为 x 、 y 、 z ，这就是点 A （或向量 $\overrightarrow{OA}$ ）的坐标来源。(The coordinates of points B , C and D on the coordinate axes are x , y and z respectively, which is the source of the coordinates of point A (or vector $\overrightarrow{OA}$).)

四、例题解析(Example Analysis)

例 1(Example 1)

在长方体 $OABC-D'A'B'C'$ 中， $OA=3$ ， $OC=4$ ， $OD'=2$ ，以 $\left\{\frac{1}{3}\overrightarrow{OA},\frac{1}{4}\overrightarrow{OC},\frac{1}{2}\overrightarrow{OD'}\right\}$ 为单位正交基底，建立空间直角坐标系 $Oxyz$ 。(As shown in Figure 1.3-6, in the cuboid $OABC-D'A'B'C'$, $OA=3$, $OC=4$, $OD'=2$. Establish a space rectangular coordinate system



$Oxyz$ with $\left\{\frac{1}{3}\overrightarrow{OA},\frac{1}{4}\overrightarrow{OC},\frac{1}{2}\overrightarrow{OD'}\right\}$ as the orthonormal basis.)

（1）写出 D' 、 C 、 A' 、 B' 四点的坐标；(www.sicyes.com) (1) Write the coordinates of the four points D' , C , A' , B' ;

（2）写出向量 $\overrightarrow{A'B'}$ 、 $\overrightarrow{B'B}$ 、 $\overrightarrow{A'C'}$ 、 $\overrightarrow{AC'}$ 的坐标。(2) Write the coordinates of the vectors $\overrightarrow{A'B'}$, $\overrightarrow{B'B}$, $\overrightarrow{A'C'}$, $\overrightarrow{AC'}$.

解：(Solution:)

（1）求四点坐标：(1) Finding the coordinates of the four points:

- 点 D' 在 z 轴上，其在 x 轴、y 轴上的射影均为原点 O ，且 $OD'=2$ ，所以 $\overrightarrow{OD'}=0i+0j+2k$ ，因此点 D' 的坐标是 $(0,0,2)$ ；(Point D' is on the z -axis, and its projections on the x -axis and y -axis are both the origin O , and $OD'=2$, so $\overrightarrow{OD'}=0i+0j+2k$, therefore the coordinate of point D' is $(0,0,2)$;)
- 点 C 在 y 轴上，其在 x 轴、z 轴上的射影均为原点 O ，且 $OC=4$ ，所以点 C 的坐标是 $(0,4,0)$ ；(Point C is on the y -axis, and its projections on the x -axis and z -axis are both the origin O , and $OC=4$, so the coordinate of point C is $(0,4,0)$;)
- 点 A' 在 x 轴、 y 轴、 z 轴上的射影分别为 A 、 O 、 D' ，它们在坐标轴上的坐标分别为 3 、 0 、 2 ，所以点 A' 的坐标是 $(3,0,2)$ ；(The projections of point A' on the x -axis, y -axis and z -axis are A , O and D' respectively, and their coordinates on the coordinate axes are 3 , 0 and 2 respectively, so the coordinate of point A' is $(3,0,2)$;)
- 点 B' 在 x 轴、 y 轴、 z 轴上的射影分别为 A 、 C 、 D' ，它们在坐

标轴上的坐标分别为 3 、 4 、 2 ，所以点 B' 的坐标是 $(3,4,2)$ 。(The projections of point B' on the x -axis, y -axis and z -axis are A , C and D' respectively, and their coordinates on the coordinate axes are 3 , 4 and 2 respectively, so the coordinate of point B' is $(3,4,2)$.)

（2）求向量坐标：(2) Finding the coordinates of the vectors:

- $\overrightarrow{A'B'} = \overrightarrow{OC}$ （长方体中$A'B' \parallel OC$且$A'B' = OC$，向量方向相同、长度相等，故向量相等），而 $\overrightarrow{OC} = 0i + 4j + 0k$ ，因此 $\overrightarrow{A'B'} = (0,4,0)$ ；($\overrightarrow{A'B'} = \overrightarrow{OC}$ (In a cuboid, $A'B' \parallel OC$ and $A'B' = OC$. Since the vectors have the same direction and equal length, they are equal).And $\overrightarrow{OC} = 0\mathbf{i} + 4\mathbf{j} + 0\mathbf{k}$, thus $\overrightarrow{A'B'} = (0,4,0)$.)
- $\overrightarrow{B'B} = -\overrightarrow{OD'}$ （ $\overrightarrow{B'B}$ 与 $\overrightarrow{OD'}$ 方向相反，长度相等），(www.sicyes.com)而 $\overrightarrow{OD'} = 0i + 0j + 2k$ ，因此 $\overrightarrow{B'B} = (0,0,-2)$ ；($\overrightarrow{B'B} = -\overrightarrow{OD'}$ ($\overrightarrow{B'B}$ and $\overrightarrow{OD'}$ are in opposite directions and equal in length), and $\overrightarrow{OD'} = 0i + 0j + 2k$, therefore $\overrightarrow{B'B} = (0,0,-2)$;)
- $\overrightarrow{A'C'} = \overrightarrow{A'D'} + \overrightarrow{D'C'}$ （向量加法的三角形法则）, 其中 $\overrightarrow{A'D'} = -3i + 0j + 0k$ ， $\overrightarrow{D'C'} = 0i + 4j + 0k$ ，因此 $\overrightarrow{A'C'} = -3i + 4j + 0k = (-3,4,0)$ ；($\overrightarrow{A'C'} = \overrightarrow{A'D'} + \overrightarrow{D'C'}$ (triangle rule of vector addition), where $\overrightarrow{A'D'} = -3i + 0j + 0k$ and $\overrightarrow{D'C'} = 0i + 4j + 0k$, therefore $\overrightarrow{A'C'} = -3i + 4j + 0k = (-3,4,0)$;)
- $\overrightarrow{AC'} = \overrightarrow{AO} + \overrightarrow{OC} + \overrightarrow{CC'}$ （向量加法的多边形法则），其中 $\overrightarrow{AO} = -3i + 0j + 0k$ ， $\overrightarrow{OC} = 0i + 4j + 0k$ ， $\overrightarrow{CC'} = 0i + 0j + 2k$ ，因此 $\overrightarrow{AC'} = -3i + 4j + 2k = (-3,4,2)$ 。($\overrightarrow{AC'} = \overrightarrow{AO} + \overrightarrow{OC} + \overrightarrow{CC'}$ (polygonal rule of vector addition), where $\overrightarrow{AO} = -3i + 0j + 0k$ ， $\overrightarrow{OC} = 0i + 4j + 0k$ ， and $\overrightarrow{CC'} = 0i + 0j +$

$2k$, therefore $\overrightarrow{AC'} = -3i + 4j + 2k = (-3,4,2)$.)

五、知识小结(Knowledge Summary)

1. 空间直角坐标系是平面直角坐标系在空间中的拓展，核心是“原点+三条两两垂直的单位坐标轴”，遵循右手定则，用于将空间中的点和向量转化为有序实数组。(A space rectangular coordinate system is an extension of the plane rectangular coordinate system in space. Its core is “origin + three mutually perpendicular unit coordinate axes”, following the right-hand rule, and it is used to convert points and vectors in space into ordered real number triples.)
2. 点的坐标由其在三条坐标轴上的射影坐标确定(www.sicyes.com)，向量的坐标与以原点为起点、该向量终点为坐标的点的坐标一致。(The coordinates of a point are determined by its projection coordinates on the three coordinate axes, and the coordinates of a vector are consistent with the coordinates of the point with the origin as the starting point and the end point of the vector as the coordinate.)
3. 空间直角坐标系的建立实现了空间几何问题的代数化，为后续通过坐标运算研究空间向量的关系、空间中的距离与角度等问题奠定了基础。(The establishment of a space rectangular coordinate system realizes the algebraization of spatial geometric problems, laying a foundation for subsequent research on the relationship between spatial vectors, distances and angles in space through coordinate operations.)